INT315 - Cluster Computing

Sales Trend Prediction using PySpark

AIM:

To predict future sales trends by analyzing historical sales data using PySpark, applying lag features, and building a Linear Regression model.

OBJECTIVES:

1. Load and process large-scale sales data using PySpark.
2. Clean and handle missing or malformed numeric data safely.
3. Create lag features (previous month and previous 3-month sales).
4. Train a Linear Regression model to predict future monthly sales.
5. Evaluate model performance using RMSE metric.
6. Visualize Actual vs Predicted sales trends.

TOOLS & TECHNOLOGIES:

- PySpark (Big data processing & ML modeling)
- Python 3.11+
- Spark MLlib
- Dataset: Superstore Sales Dataset (train.csv)
- Environment: Spark 4.0.1 (Hadoop 3)

DATASET DETAILS:

Global Superstore Sales Dataset - 4 years (Kaggle)

File: train.csv

Key Columns: Order Date, Sales, Category, Region, etc.

THEORY:

PySpark is a distributed data processing framework with MLlib for machine learning.

Linear Regression models continuous trends, using lag features like Prev_Month_Sales.

RMSE evaluates prediction accuracy (lower = better).

PROCEDURE:

1. Start Spark session
2. Load CSV dataset
3. Clean & parse dates
4. Clean 'Sales' column (remove commas, blanks, invalid data)
5. Aggregate monthly sales
6. Create lag features (1 and 3 months)
7. Train Linear Regression model
8. Evaluate using RMSE
9. Save predictions to CSV

SAMPLE OUTPUT:

Spark Session Started Successfully!

Cleaned Sales Data

Monthly Aggregated Sales

Lag Features Created

Linear Regression Model Trained

RMSE: 243.56

Predictions Saved to 'Predicted_Sales_Output'

RESULT:

The PySpark Linear Regression model successfully predicted monthly sales using lag features.

RMSE confirms acceptable model accuracy for trend prediction.

VIVA QUESTIONS:

1. What is PySpark?

   PySpark is the Python API for Apache Spark.

2. Why use lag features?

   To include time-based dependencies from previous sales.

3. What is RMSE?

   Root Mean Squared Error, evaluates model accuracy.

4. Why Linear Regression?

   Simple and effective for continuous prediction.

5. What is VectorAssembler?

   Combines multiple input columns into a single feature vector.

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Experiment: Sales Trend Prediction using PySpark